WHFNet: A Wavelet-Driven Heterogeneous Fusion Network for High-Frequency Enhanced Optical–SAR Remote Sensing Segmentation

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***Abstract*—Synergizing the visual richness of optical imagery with the structural robustness of synthetic aperture radar (SAR) is pivotal for land cover classification (LCC). However, the inherent heterogeneity and distinct data distributions between these active and passive modalities often lead to "negative transfer," where valuable modality-specific information is mixed during fusion. To overcome these limitations, this article proposes a wavelet-driven heterogeneous cross-modal fusion network (WHFNet) for coregistered optical and SAR images. Unlike normally symmetric architectures, WHFNet adopts a decoupled heterogeneous encoding strategy, leveraging the distinct inductive biases of Transformers and convolutional neural networks (CNNs) to preserve independent semantic representations. To bridge the modality gap, we introduce a cross-modal interactor (CMI) grounded in the 2-D discrete wavelet transform (DWT). This module explicitly injects high-frequency information into spatial features, enhancing the representation of details. Furthermore, a spatial–frequency fusion module (SFFM) is devised to dynamically calibrate the discrepancies between modalities via subtraction operation, while a structural consistency constraint promotes semantically aligned predictions across modalities. Extensive experiments on four benchmark datasets (Xi'an, Pohang, WHU-OPT-SAR, and PIE-RGB-SAR) demonstrate that WHFNet establishes new state-of-the-art performance. Notably, it achieves substantial accuracy gains, particularly improving mIoU by 1.25% on the cloudless WHU-OPT-SAR dataset and 0.92% on the high-precision Xi'an dataset. The code will be publicly available at https://github.com/XD-MG/WHFNet**

***Index Terms*—Land cover classification (LCC), multimodal fusion, remote sensing (RS) image, wavelet transform.**

I. Introduction

ACCURATE land cover classification (LCC) is critical for remote sensing (RS) applications yet remains challenging due to the inherent limitations of any single data source [1], [2]. The fusion of multimodal data, particularly optical and synthetic aperture radar (SAR) imagery, has emerged as a promising solution. Optical imagery often delivers ample visual information but is sensitive to atmospheric conditions, which greatly degrades feature visibility. In contrast, SAR, with its active microwave imaging capability, operates independently of weather or illumination and penetrates cloud and fog, capturing surface information. This complementarity offers significant potential gains in classification accuracy [3]. However, there are inherent differences between optical and SAR imaging modalities. Therefore, how to effectively integrate heterogeneous RS data remains a persistent challenge [4], [5], [6].

In the past studies, there have been many methods [7], [8], [9] to integrate multimodal datasets to achieve cross-modal fusion classification. Initially, modalities were processed independently with decision-level fusion [10]. With advancing research, feature-level fusion approaches have gained prominence. For instance, MCANet [11] designed a low-to-high feature fusion module to achieve spatial and semantic information integration, and SwinTFNet [12] leveraged Transformer-based global context modeling combined with cross-attention fusion to improve segmentation performance. In addition, recent studies [13], [14] have attempted to explore the structural consistency and complementarity between optical and SAR modalities. SDA [15] aligned high-level semantic features by projecting them into a shared latent space to promote semantic consistency. Despite these improvements in intermodal interactions, two major limitations persist. First, most methods lack a heterogeneous feature extraction mechanism tailored to the inherent disparities between optical and SAR data. Second, previous approaches often overlook the significance of high-frequency information, leading to an inadequate preservation of structural details such as edges and textures.

It is worth noting that convolutional neural networks (CNNs) [16], [17] tend to prioritize low-frequency components during feature extraction. Operations such as pooling and strided convolution tend to gradually diminish high-frequency details [18]. This results in the critical high-frequency information, essential for distinguishing ambiguous land cover categories like road edges and water boundaries, being progressively lost, ultimately compromising classification

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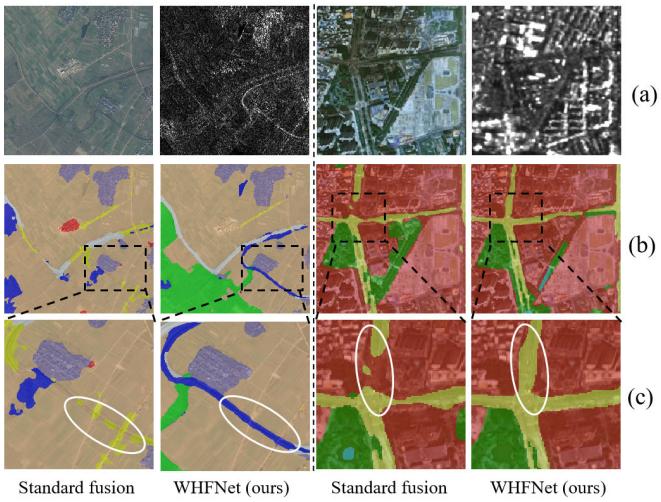


Fig. 1. (a) Image pairs. (b) Predictions of models. (c) Enlarged image of (b). In (b) and (c), the first and third columns show results from previous feature fusion methods SA-Gate, and it exhibits discontinuous classification boundaries. The remaining columns are from the proposed method WHFNet.

accuracy. As shown in Fig. 1, existing fusion methods often yield blurred classification boundaries with loss of details. Given the demonstrated capability of wavelet transforms in extracting textual and high-frequency information from RS imagery [19], [20], we propose to incorporate wavelet transforms into the fusion pipeline to enhance the retention and utilization of high-frequency features, thereby improving the discriminative power of cross-modal fusion.

Additionally, optical sensors capture reflected light across spectral bands, providing color and texture information, while SAR actively acquires microwave backscatter signals, reflecting surface roughness, moisture, and structural geometry [21], and its features are often directionally sensitive [22]. Therefore, a heterogeneous fusion framework is proposed to separate the differential representations of heterogeneous data. Building on these insights, we aim to develop a heterogeneous fusion network that can leverage unique and complementary characteristics of optical and SAR modalities to improve LCC accuracy.

In this article, we propose a wavelet-driven heterogeneous cross-modal fusion network (WHFNet) for LCC. Considering the distinct imaging mechanisms of the two modalities, WHFNet adopts a Transformer–CNN heterogeneous architecture. Specifically, to handle the complex visual information in optical images, the Swin Transformer is employed to capture global semantic contexts and long-range dependencies. The ConvNeXt network is deployed for SAR data, which leverages inherent local inductive biases to effectively model local scattering patterns. To compensate for the loss of fine details during deep encoding, high-frequency components for both modalities are extracted through Haar wavelet decomposition, serving as structural priors for the subsequent fusion stage. A cross-modal interactor (CMI) is further developed. It has two important functions. First, it processes the high-frequency features generated by the wavelet transform through embedded texture-aware blocks. Second, it dynamically captures the complementary features of spatial and frequency domains through the spatial–frequency fusion module (SFFM) to improve the quality of feature fusion. Moreover, a structural consistency loss (SCL) ensures that the network learns semantically consistent interpretations across modalities. These components enable WHFNet to leverage each modality's features in a heterogeneous manner, leading to the precise classification results illustrated in Fig. 1.

The main contributions of this article can be summarized as follows.

1) A heterogeneous cross-modal fusion parallel network WHFNet is proposed, which explicitly decouples the feature extraction of optical and SAR modalities. This architecture preserves modality-specific characteristics and supports more comprehensive interactions.
2) To explicitly leverage high-frequency information, a CMI for stage-wise feature enhancement and fusion is introduced. By integrating high-frequency components based on Haar wavelet decomposition with a unified spatial–frequency fusion mechanism, CMI provides richer structural information and enhances the overall quality of feature alignment.
3) An SFFM is developed. It captures the complementary gap between spatial and frequency domains via subtraction operation. Through residual-based calibration, SFFM dynamically calibrates spatial features and produces more discriminative fused representations conducive to accurate LCC.

The overall structure of this article is as follows. In Section II, related work is described, and Section III details the principles of the proposed framework and its components. Section IV demonstrates the effectiveness of WHFNet through quantitative and qualitative experimental analysis. Finally, Section V summarizes this article and presents future considerations.

II. Related Work

A. Land Use and LCC

LCC in RS images often faces challenges [23]. In recent years, some researchers have applied machine learning algorithms to LCC tasks, which has attracted extensive attention [24], [25]. It primarily relies on feature selection and classifier design, including support vector machine (SVM), random forest (RF), Naive Bayesian classifier, and so on [26], [27], [28]. The SVM is considered to be effective in processing high-dimensional RS images, surpassing the decision tree classifier in theory and stability [29]. However, when solving large-scale image data, the memory resource and time consumption will be greatly increased. Gislason et al. [27] proved that the RF classifier can also maintain good classification performance in complex regional scenes. However, the classifier design is difficult to capture the spatial correlation of ground objects, resulting in a decrease in the edge accuracy.

Deep learning networks have been widely adopted in various RS tasks. For example, Pan et al. [30] introduced a CNN-based multispectral LiDAR LCC framework and

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its optimal parameter settings. With the strong segmentation performance of UNet [31] demonstrated in medical images, the encoder–decoder structure has also been extensively applied to RS image classification. Ghosh et al. [32] proposed dilated stacked UNets to process high-resolution optical imagery, while ResUNet-a [33] employed UNet as its backbone and integrated residual connections, atrous convolutions, and pyramid scene parsing pooling for pixel-level object classification. More recently, inspired by advancements in computer vision architectures (ViT [34], Swin Transformer [35], [36], ConvNeXt [37], [38], and so on), researchers have begun exploring the potential of Transformer models for RS applications, achieving remarkable progress [39], [40]. For example, Liu et al. [41] proposed the global–local network structure (GLNS) to address the interference of coherent speckle [42] in SAR image classification, which learns local and global features by fusing a lightweight CNN and ViT network. Similarly, the ST-U-shaped network [43] embedded the Swin Transformer into the UNet framework, resulting in significant performance gains on the Potsdam and Vaihingen datasets.

B. Multimodal Fusion

Despite great advances in LCC tasks, single RS data remain limited by large intraclass variance and high interclass similarity [3]. Multimodal fusion methods offer improved accuracy through complementary feature representations [44], [45]. Current multimodal fusion approaches in RS primarily differ in their fusion strategies, which can be categorized into three groups: early fusion, intermediate fusion, and late fusion. Early fusion combines raw inputs to preserve information, while late fusion merges decisions after classification, limiting cross-modal interaction. In contrast, intermediate fusion provides greater flexibility by implementing hierarchical feature integration strategies that enhance multimodal feature representation. Ma et al. [9] proposed CMFNet, a cross-modal fusion network that employs multiscale skip connections to mitigate information loss during decoding. Building on this foundation, CMGFNet [46] designed a targeted fusion scheme for RGB and DSM data, further improving cross-modal feature alignment. SDIR [47] provided theoretical insights by modeling similarity and dissimilarity structures within multimodal data via k-nearest and k-farthest neighbor graphs. Beyond spatial multimodality, STA-AgriNet [48] extends fusion to the temporal domain, employing spatiotemporal attention to integrate multisensor, multitemporal satellite image time series for crop-type mapping, highlighting the value of temporal context in multimodal analysis.

In optical–SAR fusion research, recent approaches [49], [50] have focused on modeling semantic correlations to enhance complementary feature learning across heterogeneous modalities. DDHRNet [51] adopts a dual-stream high-resolution architecture with squeeze-and-excitation modules to achieve deep cross-layer fusion. Liu et al. [52] further develop a joint learning strategy that aggregates optical–SAR features within a shared semantic space. Additional advances include ASANet [53], which adopts asymmetric semantic alignment to enhance optical–SAR LCC accuracy, emphasizing the need for modality-specific feature calibration. Gao et al. [54] introduced DEN, a dual-encoder network for SAR-optical fusion, demonstrating the benefits of structural asymmetry and adaptive fusion in handling different modalities. Despite their effectiveness, current methods still struggle with nonlinear radiation discrepancies between optical and SAR sensors. Such inherent differences weaken cross-modal complementarity and may cause valuable modality-specific information to be suppressed during aggregation. To overcome these limitations, our work proposes a new solution: 1) we introduce frequency-domain analysis to separate noise components from semantic features and 2) develop a heterogeneous backbone network that synergistically exploits spatial and frequency information through an innovative fusion strategy.

C. Frequency-Domain Fusion

Frequency-domain information has shown unique advantages in RS image processing [14]. Early works used the Fourier transform to achieve global spectral analysis. For example, in [55], the improved local Fourier histograms texture feature classification technique was applied to multibeam echosounder (MBES) data. Singh and Datcu [56] proved that the fractional Fourier transform (FrFT) is suitable for SAR image classification tasks. Subsequently, the introduction of the wavelet transform promoted the development of frequency-domain analysis [57], [58]. A classification model based on the texture energy of wavelet coefficients was introduced by [59], which effectively removed the speckle noise interference in SAR images. Li et al. [60] uses 2-D oriented wavelet transform to increase the gain of transform coding. Alonso et al. [61] present an unsupervised edge enhancement algorithm for SAR based on a combination of wavelet coefficients at different scales.

In terms of feature fusion, the introduction of a series of frequency-domain techniques has promoted diversified feature learning. Cheng et al. [62] used the wavelet transform to obtain the intensity components of multispectral images and panchromatic images. Jia et al. [63] used the complementary information of two difference images fused by the multiple wavelet kernels for change detection. In [64], the wavelet transform was used to extract decomposition features and calculate the similarity coefficients of different source components. Some works [65], [66] have shown that the high-frequency subbands generated by the wavelet transform contain rich texture variation information in the original image. Moreover, rotation invariance and the multiscale approach of these wavelets make the feature extraction robust. A high-frequency attention-guided Siamese network (HFA-Net) [67] enhanced the high-frequency information of buildings by designing the high-frequency attention block (HFAB) module. In this work, we introduce the wavelet transform to extract high-frequency information and propose a novel heterogeneous fusion network (WHFNet), which effectively combines spatial and frequency features to enhance feature representation.

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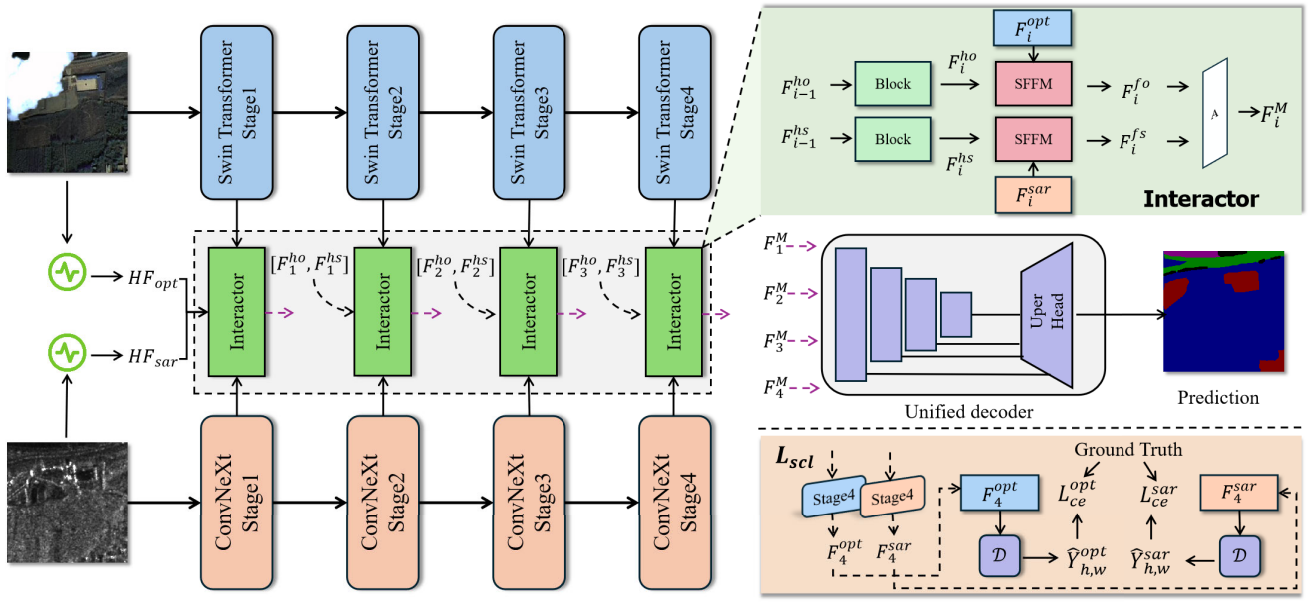


Fig. 2. Overview of the proposed network WHFNet. In the structure of CMI, "A" means the addition operation. In the design of SCL, "$\mathcal{D}$" denotes the auxiliary decoder.

III. Methodology

This section details the proposed WHFNet, as illustrated in Fig. 2. We begin by outlining its overall architecture and the data flow pipeline. Following this, we describe how the wavelet transform is utilized to explicitly extract high-frequency priors. Subsequently, the CMI is introduced, which integrates texture-aware blocks and the SFFM at multiple stages to achieve progressive feature refinement. The SFFM is then analyzed in detail, focusing on its ability to combine complementary features from different domains. Finally, the function of the SCL is described, which enforces the network to learn deep semantic representations that are modality-invariant during optimization.

A. Overview of Our Framework

To address the heterogeneous characteristics of optical and SAR images, distinct feature extraction pathways were designed. The optical branch employs Swin Transformer, while the SAR branch utilizes ConvNeXt. Subsequently, the Haar wavelet transform is used to extract high-frequency components from each modality. This transform not only fully exploits the high-frequency details (such as edges and textures) in the images but also effectively alleviates the interference of nonlinear intensity variations in SAR images, providing complementary information for the backbone branches. For effective fusion, we designed a CMI to unified fusion of complementary features from both modalities. It contains texture-aware blocks to extract high-frequency features and the SFFM to fuse cross-domain features. The fused features from different stages are then aggregated and input into the cascade decoder UperHead. The final segmentation map is obtained by the upsampling operation. Additionally, we apply SCL to the fourth stage backbone features.

Let $\mathbf{I}_{\text{opt}}$ and $\mathbf{I}_{\text{sar}}$ denote the input optical and SAR images, respectively. The feature maps extracted by the backbone network at different stages are represented as $\mathbf{F}_i^{\text{opt}}$ and $\mathbf{F}_i^{\text{sar}}$ ($i = 1, 2, 3, 4$), where the superscripts distinguish between the two modalities. The spatial dimensions of the feature maps at each stage follow:

$$\begin{aligned}\mathbf{F}_i^{\text{opt}} &= \frac{H}{2^{i+1}} \times \frac{W}{2^{i+1}} \times c_i \\ \mathbf{F}_i^{\text{sar}} &= \frac{H}{2^{i+1}} \times \frac{W}{2^{i+1}} \times c_i \end{aligned} \tag{1}$$

where H and W represent the height and width of the input image, respectively, and c_i denotes the number of out channels in the ith stage. Correspondingly, the high-frequency component extraction by the texture-aware block is denoted as $\mathbf{F}_i^{ho}$ and $\mathbf{F}_i^{hs}$, where o denotes the optical and s represents the SAR. The dimensions of these high-frequency feature maps can be expressed as follows:

$$\begin{aligned}\mathbf{F}_i^{ho} &= \frac{H}{2^{i+1}} \times \frac{W}{2^{i+1}} \times c_i' \\ \mathbf{F}_i^{hs} &= \frac{H}{2^{i+1}} \times \frac{W}{2^{i+1}} \times c_i'. \end{aligned} \tag{2}$$

In this article, we maintain $c_i' = c_i$ ($i = 1, 2, 3, 4$) across all stages to ensure dimensional consistency between the backbone and frequency features. To maintain the structural integrity of two modalities, the CMI integrates spatial features ($\mathbf{F}_i^{\text{opt}}$ and $\mathbf{F}_i^{\text{sar}}$) and frequency features ($\mathbf{F}_i^{ho}$ and $\mathbf{F}_i^{hs}$) at each stage. The process is formally defined as

$$\mathbf{F}_i^M, \left[\mathbf{F}_i^{ho}, \mathbf{F}_i^{hs}\right] = \mathcal{F}_{\text{CMI}}\left(\mathbf{F}_i^{\text{opt}}, \mathbf{F}_{i-1}^{ho}, \mathbf{F}_i^{\text{sar}}, \mathbf{F}_{i-1}^{\text{so}}\right) \tag{3}$$

where $\mathcal{F}_{\text{CMI}}$ denotes the cross-modal interactor. The implementation detail of CMI is presented in Section III-C. Particularly, the initial input are denoted as $\mathbf{F}_0^{ho} = \text{HF}_{\text{opt}}, \mathbf{F}_0^{hs} = \text{HF}_{\text{sar}}$. The output feature map $\mathbf{F}_i^M$ is obtained by element-wise addition of the enhanced optical and SAR branch features,

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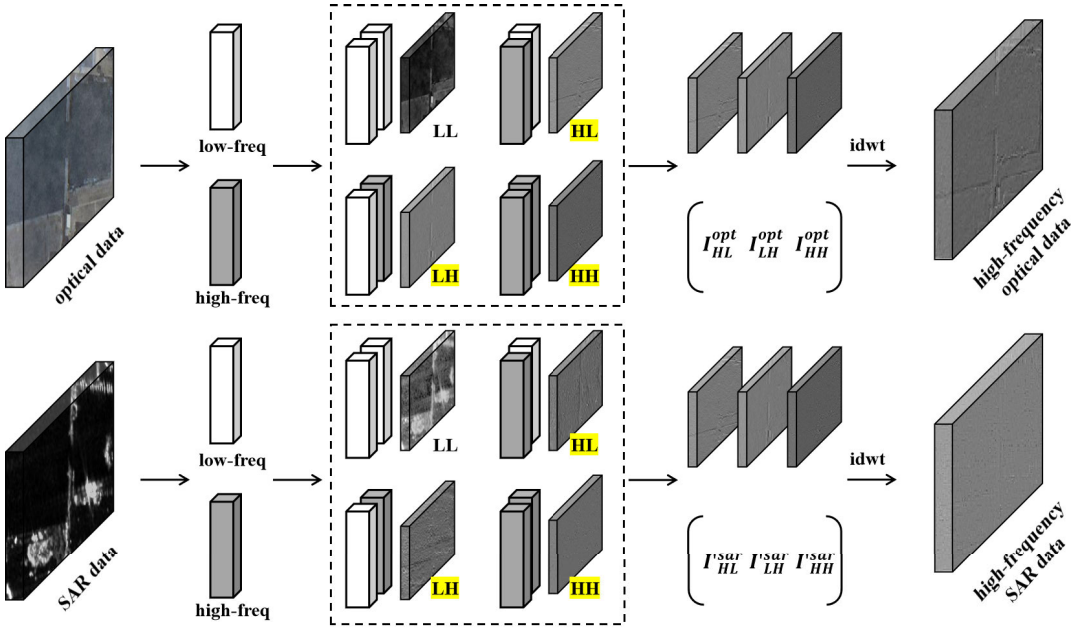


Fig. 3. Process of high-frequency images generation from optical and SAR images via 2-D wavelet transform.

and it maintains the original channel dimension c_i of the ith stage. The resulting multiscale fused features, denoted as $\{\mathbf{F}_1^M, \mathbf{F}_2^M, \mathbf{F}_3^M, \mathbf{F}_4^M\}$, are subsequently aggregated and decoded by the UPerHead. The decoder integrates semantic information across different scales, enabling precise prediction of results with the original size. The UPerHead is optimized using the standard cross-entropy loss. Furthermore, the SCL is detailed in Section III-E.

B. High-Frequency Image Based on Wavelet Transform

The wavelet transform is an effective tool for decomposing spatial and frequency information, playing a crucial role in characterizing texture details [57]. Consequently, we introduce the discrete wavelet transform (DWT) into the network to obtain high-frequency components.

The 2-D wavelet decomposition yields a set of subband images, consisting of low-frequency components containing global semantic information and detail image sequences containing higher frequency information. Taking optical image as an example, the original input image $\mathbf{I}$ is transformed by 2-D wavelet transform to obtain four components: $\mathbf{I}_{\mathrm{LL}}, \mathbf{I}_{\mathrm{LH}}, \mathbf{I}_{\mathrm{HL}}$, and $\mathbf{I}_{\mathrm{HH}}$, as shown in Fig. 3. The expression is as follows:

$$\mathbf{I}_{\mathrm{LL}}, (\mathbf{I}_{\mathrm{LH}}, \mathbf{I}_{\mathrm{HL}}, \mathbf{I}_{\mathrm{HH}}) = \mathrm{DWT}(\mathbf{I}). \tag{4}$$

Each component represents frequency responses along different directions, $\mathbf{I}_{\mathrm{LL}}$ represents the low-frequency information of the image, the $\mathbf{I}_{\mathrm{LH}}$ subband represents the high-frequency changes in the image in the vertical direction, $\mathbf{I}_{\mathrm{HL}}$ captures high-frequency variations along the horizontal direction, and $\mathbf{I}_{\mathrm{HH}}$ contains diagonal high-frequency components.

Due to the difference in imaging mechanism, the amount of information conveyed by optical images and SAR images is also different. Optical images are typically corrupted by additive noise, while SAR images, which rely on coherent processing and synthetic aperture techniques for high-resolution imaging, are primarily affected by multiplicative speckle noise [68]. Therefore, the modalities can be formally defined as

$$\begin{aligned} \mathbf{I}_{\mathrm{opt}} &= \mathbf{F}_l + \mathbf{F}_h + \mathbf{N}_l + \mathbf{N}_h \\ \mathbf{I}_{\mathrm{sar}} &= (\mathbf{F}_{sl} + \mathbf{F}_{sh}) * \mathbf{N}_s \end{aligned} \tag{5}$$

where $\mathbf{F}_l$ is the low-frequency structural feature; $\mathbf{F}_h$ is the high-frequency detail feature; and $\mathbf{N}_l$ and $\mathbf{N}_h$ represent the low- and high-frequency noise in the optical image, respectively. $\mathbf{N}_s$ is the multiplicative speckle noise, typically following an exponential or gamma distribution. In addition, we define $\mathbf{LF}$ and $\mathbf{HF}$ as the low-frequency image and high-frequency image obtained after applying the wavelet transform. Thus, the optical image decomposition can be represented as follows:

$$\begin{aligned} \mathbf{L}_{\mathrm{opt}}, \mathbf{H}_{\mathrm{opt}} &= \mathrm{DWT}\left(\mathbf{I}_{\mathrm{opt}}\right) \\ \mathbf{L}_{\mathrm{opt}}, \mathbf{H}_{\mathrm{opt}} &= \mathbf{F}_l + \mathbf{N}_l, \mathbf{F}_h + \mathbf{N}_h \\ \mathbf{LF}_{\mathrm{opt}}, \mathbf{HF}_{\mathrm{opt}} &= \mathrm{iDWT}\left(\mathbf{L}_{\mathrm{opt}}\right), \mathrm{iDWT}\left(\mathbf{H}_{\mathrm{opt}}\right). \end{aligned} \tag{6}$$

To simplify the progress in SAR image decomposition, a logarithmic transformation is employed to convert the original multiplicative speckle noise into an additive noise model. This transformation enables the extraction of frequency feature images, denoted as $\mathbf{LF}_{\mathrm{sar}}, \mathbf{HF}_{\mathrm{sar}}$

$$\begin{aligned} \mathbf{I}'_{\mathrm{sar}} &= \log\left(\mathbf{I}_{\mathrm{sar}}\right) = \log\left(\mathbf{F}_{sl} + \mathbf{F}_{sh}\right) + \log(N) \\ \mathbf{L}_{\mathrm{sar}}, \mathbf{H}_{\mathrm{sar}} &= \mathrm{DWT}\left(\mathbf{I}'_{\mathrm{sar}}\right) \\ \mathbf{LF}_{\mathrm{sar}}, \mathbf{HF}_{\mathrm{sar}} &= \mathrm{iDWT}\left(\mathbf{L}_{\mathrm{sar}}\right) \mathrm{iDWT}\left(\mathbf{H}_{\mathrm{sar}}\right). \end{aligned} \tag{7}$$

As shown in Fig. 3, the high-frequency components of both optical and SAR images are displayed. These are then used to enhance feature representation within the CMI.

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C. Cross-Modal Interactor

The heterogeneous encoders provide modality-specific representations. However, directly merging these divergent features often leads to suboptimal fusion due to their incompatible characteristics. To bridge this gap and exploit the complementary information between frequency and spatial features, we introduce a unified interactor, CMI, which serves as the core fusion unit at each stage of our network. This module is designed to perform effective alignment and fusion between optical and SAR features.

As illustrated in Fig. 2, each interactor accepts two types of inputs: spatial features from the encoders ($\mathbf{F}_i^{\text{opt}}, \mathbf{F}_i^{\text{sar}}$) and high-frequency features from the previous stage ($\mathbf{F}_{i-1}^{ho}, \mathbf{F}_{i-1}^{hs}$). The feature interaction is conducted in two steps: frequency feature enhancement and feature fusion. The texture-aware block corresponding to each stage in CMI shares an identical architectural design. The output feature maps are formulated as follows:

$$\begin{aligned}\mathbf{F}_1^{\text{tab}} &= \text{RELU}\left(\text{BN}\left(\text{Conv}_1\left(\mathbf{I}_{\text{HF}}\right)\right)\right)\\ \mathbf{F}_i^{\text{tab}} &= \text{MAP}\left(\text{RELU}\left(\text{BN}\left(\text{Conv}_2\left(\mathbf{F}_{i-1}^{\text{tab}}\right)\right)\right)\right), \quad i = 2,3,4\end{aligned} \tag{8}$$

where $\mathbf{I}_{\text{HF}}$ denotes the high-frequency image, which can be either an optical image or an SAR image. The notation $\mathbf{F}_i^{\text{tab}}$ denoted as $\mathbf{F}_i^{ho}$ or $\mathbf{F}_i^{hs}$ corresponding to the high-frequency features from the optical and SAR modalities, respectively ($i = 1,2,3,4$). Specifically, Conv_1 is a convolutional layer with a kernel size of 4, and Conv_2 employs a kernel size of 3. The operator MAP represents the max pooling operation, which performs downsampling of the feature maps by a factor of 2 in each spatial dimension. Subsequently, the high-frequency features and spatial features are fused by SFFM, which can be summarized as follows:

$$\begin{aligned}\mathbf{F}_i^{fo} &= \mathcal{F}_{\text{SFFM}}\left(\mathbf{F}_i^{\text{opt}}, \mathbf{F}_i^{ho}\right)\\ \mathbf{F}_i^{fs} &= \mathcal{F}_{\text{SFFM}}\left(\mathbf{F}_i^{\text{sar}}, \mathbf{F}_i^{hs}\right)\end{aligned} \tag{9}$$

where $\mathbf{F}_i^{fo}$ and $\mathbf{F}_i^{fs}$ denote the fused optical and SAR features, respectively, at stage i. This fusion mechanism preserves the inherent spatial characteristics of each imaging modality while facilitating complementary feature aggregation. Following the fusion operations defined in (10), the output feature map of the corresponding stage is obtained by element-wise addition of the enhanced optical and SAR branch features

$$\mathbf{F}_i^M = \mathbf{F}_i^{fo} + \mathbf{F}_i^{fs}. \tag{10}$$

By preserving and processing these high-frequency components in a parallel way, the model implements more comprehensive feature interaction.

D. Spatial–Frequency Fusion Module

Features derived from the texture-aware block capture rich high-frequency statistics, whereas backbone features preserve the spatial semantics of the image. To fully synergize these spatial domain features with frequency-domain representations and maximize the complementarity of multimodal cues, we design the SFFM.

As depicted in Fig. 4, the SFFM receives two distinct inputs: spatial features $\mathbf{F}_{\text{spa}}$ and high-frequency features $\mathbf{F}_{\text{fre}}$. Taking the SAR modality as an example, we denote the spatial features as $\mathbf{F}_i^{\text{sar}}$ and the frequency features as $\mathbf{F}_i^{hs}$. The dual-domain fusion process unfolds in three key steps.

1) *Feature Projection and Difference Representation:* To ensure the consistency of scale and channel, both $\mathbf{F}_i^{\text{sar}}$ and $\mathbf{F}_i^{hs}$ are first passed through independent projection function $\phi(\cdot)$. Then, the subtraction operation is performed to capture the differences between spatial and frequency features to obtain the variable $\mathbf{Z}_i^{\text{sar}}$. The calculation process is as follows:

$$\begin{aligned}\mathbf{X}_i^{\text{sar}} &= \phi\left(\mathbf{F}_i^{\text{sar}}\right)\\ \mathbf{X}_i^{hs} &= \phi\left(\mathbf{F}_i^{hs}\right)\\ \mathbf{Z}_i^{\text{sar}} &= \mathbf{X}_i^{\text{sar}} - \mathbf{X}_i^{hs}\end{aligned} \tag{11}$$

where $i = (1,2,3,4)$ represents the four stages in the network and $\phi(\cdot)$ denotes a channel matching and projection function (typically implemented via 1×1 convolution). The different feature $\mathbf{Z}_i^{\text{sar}}$ emphasizes regions where the frequency block provides complementary fine details.

2) *Attention-Guided Frequency Feature Enhancement:* The difference feature is processed by the spatial attention module $\mathcal{F}_{\text{SA}}$ to generate an attention map highlighting significant cross-domain discrepancies. This attention map is then used to guide the weighting of the frequency features

$$\tilde{\mathbf{F}}_i^{hs} = \mathcal{F}_{\text{SA}}\left(\mathbf{Z}_i^{\text{sar}}\right) \otimes \mathbf{X}_i^{hs}. \tag{12}$$

3) *Spatial Enhancement and Feature Fusion:* The weighted frequency features $\tilde{\mathbf{F}}_i^{hs}$ obtained in the previous step are used to enhance the spatial features. To obtain a richer feature representation, the calibrated spatial features $\tilde{\mathbf{F}}_i^{s}$ are concatenated with the frequency features $\mathbf{X}_i^{hs}$, and the features are adjusted according to the operation of the following equation:

$$\begin{aligned}\tilde{\mathbf{F}}_i^{s} &= \tilde{\mathbf{F}}_i^{hs} + \mathbf{X}_i^{\text{sar}}\\ \mathbf{O}_i^{\text{sar}} &= \text{Conv}_4\left(\text{BN}\left(\text{Conv}_3\left(\text{Concat}\left(\tilde{\mathbf{F}}_i^{s}, \mathbf{X}_i^{hs}\right)\right)\right)\right).\end{aligned} \tag{13}$$

Finally, a residual connection is applied to produce the final cross-domain fused feature $\mathbf{F}_i^{fs}$

$$\mathbf{F}_i^{fs} = \mathbf{O}_i^{\text{sar}} + \mathbf{X}_i^{\text{sar}} \tag{14}$$

where Conv_3 denotes a dilated convolution and Conv_4 is a 3×3 convolution.

E. Structural Consistency Loss

Several studies [47], [69], [70] have investigated the structural consistency between optical and SAR images. To ensure that each modality stream within WHFNet maintains its specificity, we introduce an SCL. By enforcing supervision on the independent segmentation performance of each branch, the network is encouraged to preserve robust semantic feature extraction capabilities for individual modalities.

Specifically, the feature maps $\mathbf{F}_4^{\text{opt}}$ and $\mathbf{F}_4^{\text{sar}}$, derived from the final stage of the backbones, are fed into lightweight

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TABLE I
SPECIFICATIONS OF THE DATASETS USED IN EXPERIMENTS

Dataset	Resolution (m)	Input Size (pixels)	Categories	Optical data source	SAR data source
Xi'an [51]	1 m	128×128	5	GF2	GF3
Pohang [51]	1 m	256×256	5	GF2	GF3
PIE-RGB-SAR [53]	3 m	256×256	6	Google Earth	GF3
WHU-OPT-SAR [11]	5 m	512×512	7	GF-1	GF3

decoders. This process yields two maps, $\hat{\mathbf{Y}}^{\text{opt}}$ and $\hat{\mathbf{Y}}^{\text{sar}}$, which share the same resolution as the original input. To quantify the divergence between these predictions and the ground truth $\mathbf{Y}$, the cross-entropy loss is employed as the supervisory constraint

$$L_{\text{ce}}^{\text{opt}} = -\frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} \mathbf{Y}_{h,w} \log \left(\hat{\mathbf{Y}}_{h,w}^{\text{opt}} \right)$$
$$L_{\text{ce}}^{\text{sar}} = -\frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} \mathbf{Y}_{h,w} \log \left(\hat{\mathbf{Y}}_{h,w}^{\text{sar}} \right) \tag{15}$$

and the SCL L_{scl} is defined as

$$L_{\text{scl}} = \alpha L_{\text{ce}}^{\text{opt}} + \beta L_{\text{ce}}^{\text{sar}}. \tag{16}$$

In our implementation, the weighting coefficients α and β are both set to 0.3. L_{scl} serves as an independent supervisor for each modality branch, compelling them to focus on their respective spatial structural features. This regularization strategy prevents the network from becoming overly reliant on the cross-modal fusion mechanism between the spatial and frequency domains. Consequently, it mitigates the risk of semantic drift during heterogeneous feature interaction and stabilizes the model's output, ensuring robustness even when a single modality is disturbed.

IV. EXPERIMENTS

A. Datasets

To comprehensively validate the effectiveness and generalization ability of the proposed WHFNet, extensive experiments are conducted on four representative multimodal RS datasets: Xi'an, Pohang, WHU-OPT-SAR, and PIE-RGB-SAR. These datasets cover diverse geographic regions and land cover types, offering a robust benchmark for evaluating multimodal feature fusion strategies in RS semantic segmentation. An overview of all datasets used is shown in Table I.

1) Xi'an Dataset: Ren et al. [51] provided multiple multimodal datasets, among which the data in the Xi'an and Pohang regions both contain cloudless and clouded image pairs. The corresponding optical images are obtained from GF2, and the SAR images are from GF3 (slider spotlight mode). In order to verify the performance of WHFNet in LCC in complex scenes, we select the clouded image pairs for experiments. Among them, the Xi'an dataset is covered by thick clouds, and the original image resolution is 2304×3804. During the training process, the image is cropped into 128×128 blocks with a step size of 64. The dataset is partitioned equally into training and test sets, with each subset containing 1061 samples, resulting in a 1:1 split ratio.

2) Pohang Dataset: Acquired over the coastal city of Pohang, South Korea, the dataset contains 6173 image pairs. Similar to the Xi'an dataset, we select the clouded image pair for experiments. Pohang dataset simulates the interference effect under fog or smoke conditions using the GNU Image Manipulation Program software. In addition, its original image covers a wider range than the Xi'an dataset, with a size of 10240×9216. During the training process, a sliding window of size 256×256 was used for cropping, and the step size was set to 128. Finally, 3087 training images and 3086 verification images were obtained. Both the Xi'an and Pohang datasets are labeled with five types of categories: building, road, farmland, water, and greenery.

3) WHU-OPT-SAR Dataset: Li et al. [11] made significant contributions to multisource image fusion classification. They proposed a large optical–SAR land use classification dataset covering about 50 000 km^2 in Hubei Province, consisting of 100 paired optical and SAR images. The size of each image is 5556×3704. According to the distribution of test and training image IDs published by MCANet, we cropped the original image into 512×512 (nonoverlapping) image blocks and finally obtained 7040 images for training and 1760 images for testing. The ground classification scene of this dataset is complex, covering different terrains and vegetation areas, involving the following seven categories: farmland, forest, city, village, water, road, and others.

4) PIE-RGB-SAR Dataset: The PIE dataset is a new dataset proposed by Zhang et al. [53] in 2024. The research area is a part of the Pearl River Delta, China. The optical and SAR images are from Google satellite with a resolution of approximately 0.5 m and GF3 satellite (ultrafine stripe mode) with a spatial resolution of 3 m, respectively. The final aligned image resolution is 3 m. In this article, the distribution of the training set and the test set strictly follows ASANet. This dataset contains six categories: city, road, water, forest, farmland, and others.

To further elucidate the representational diversity of semantic categories across the datasets, we visualize their distribution in Fig. 5. A notable observation is the significant variance in the number of labeled samples per class among the four datasets. Specifically, the Pohang and Xi'an datasets exhibit a high concentration of samples in the building, greenery, and farmland categories. In contrast, the PIE-RGB-SAR dataset is predominantly characterized by city and forest categories, whereas the WHU-OPT-SAR dataset places a greater emphasis on water and farmland. This inherent class imbalance within and across datasets presents considerable challenges for model generalization and motivates the design of robust multimodal fusion strategies.

B. Implementation Details

This work uses the MMSegmentation framework to build the network, utilizing the AdamW optimizer with a learning rate of $1e^{-}4$. In order to quantitatively evaluate the performance of WHFNet in the optical–SAR fusion LCC task, we selected eight representative algorithms for comparison. Among these, UperNet is a single-source LCC baseline, while Segformer operates on synthetic data, where optical and SAR

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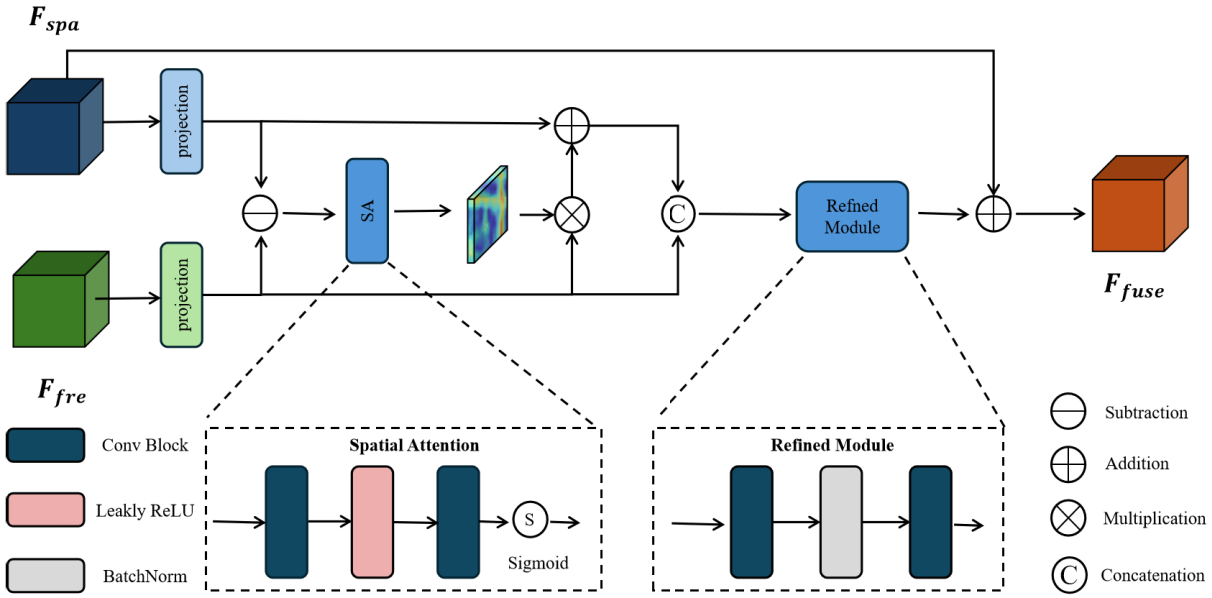


Fig. 4. Architecture of SFFM.

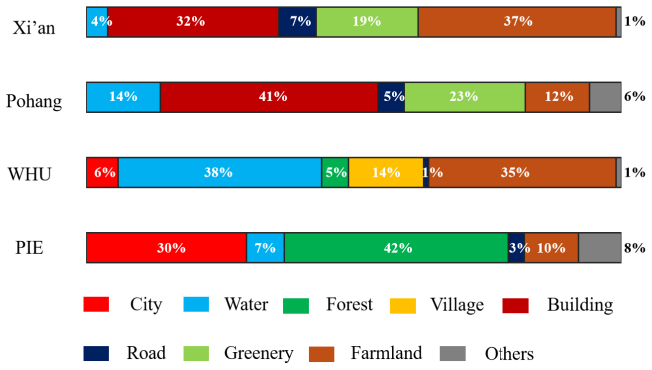


Fig. 5. Category distribution of Xi'an, Pohang, WHU-OPT-SAR, and PIE-RGB-SAR datasets.

TABLE II
HYPERPARAMETER SETTINGS FOR DIFFERENT METHODS

Model	Year	Optimizer	Lr	Momentum	Weight-decay
UperNet [71]	2018	AdamW	0.0001	(0.9, 0.999)	0.05
SA-Gate [72]	2020	SGD	0.01	0.9	0.0005
SegFormer [73]	2021	SGD	0.01	0.9	0.0005
CMFNet [9]	2022	SGD	0.01	0.9	0.0005
MCANet [11]	2022	Adam	0.001	(0.9, 0.999)	0.0001
CMX [74]	2023	AdamW	0.00006	(0.9, 0.999)	0.01
FTransUNet [75]	2024	SGD	0.01	0.9	0.0005
ASANet [53]	2024	AdamW	0.0001	(0.9, 0.999)	0.05
WHFNet	2025	AdamW	0.0001	(0.9, 0.999)	0.05

data are concatenated along the channel dimension as input. The remaining six methods represent state-of-the-art multisource RS image fusion classification algorithms. Detailed configurations of the optimizer and learning rate for each method are provided in Table II.

Given the varying scales of the selected datasets, we mitigate overfitting by adjusting the number of training iterations proportionally to the size of each training set. Specifically, the Xi'an, Pohang, WHU-OPT-SAR, and PIE-RGB-SAR datasets are trained for 15k, 40k, 80k, and 30k iterations, respectively. To maintain experimental consistency, all experiments are conducted on a platform equipped with an NVIDIA GeForce RTX 4090 (24GB) GPU, utilizing two GPUs for distributed training. A batch size of 4 is configured per GPU, and the experiments are performed on an Ubuntu 22.04 Linux system. During training, we employ data augmentation strategies including random cropping and random scaling, and a test-time augmentation strategy is utilized for evaluation.

Model performance is comprehensively assessed using the following metrics: mean intersection over union (mIoU), mean accuracy (mAcc), Kappa coefficient, overall accuracy (OA), mean Dice coefficient (mDice), per-category IoU, and per-category accuracy.

C. Experiment Performance

In this section, we compare the classification performance of the proposed WHFNet with several representative methods, including UperNet, SegFormer, SA-Gate, CMFNet, MCANet, CMX, FTransUNet, and ASANet. For UperNet, two experimental configurations are established: one leverages optical imagery (with all datasets standardized to three-channel RGB inputs) and employs the Swin Transformer as its feature extractor; the other utilizes single-source SAR images with ConvNeXt for feature extraction. SegFormer, on the other hand, processes concatenated optical and SAR data as input. SA-Gate and CMX are specialized for cross-modal fusion tasks, while the remaining models represent prominent approaches within the domain of multisource RS semantic segmentation.

1) Xi'an Dataset: Table III presents the quantitative evaluation results of all compared methods. From the overall metrics, WHFNet outperforms all algorithms by a substantial margin,

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TABLE III
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON XI'AN DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Method	Modality		Category IoU					Category Acc					Metric		
	OPT	SAR	Building	Road	Farmland	Greenery	Water	Building	Road	Farmland	Greenery	Water	mIoU	OA	Kappa
UperNet	✓		89.68	70.16	93.40	83.68	88.90	94.50	79.75	96.62	92.29	94.13	85.16	93.78	91.28
UperNet		✓	86.31	57.99	88.03	73.44	88.17	94.22	66.82	94.63	83.96	91.10	78.79	90.27	86.28
SegFormer	✓	✓	93.84	75.81	96.36	91.27	95.74	97.04	87.39	97.89	94.92	98.69	90.60	96.31	94.83
SA-Gate	✓	✓	94.13	77.30	96.66	91.54	95.39	95.69	86.41	98.86	97.27	95.97	91.00	96.53	95.12
CMFNet	✓	✓	82.45	42.57	86.75	68.38	83.47	92.23	50.86	93.57	81.86	91.64	72.73	87.56	82.46
MCANet	✓	✓	91.70	73.48	95.16	87.75	95.21	94.49	82.95	98.47	94.18	98.31	88.66	95.24	93.31
CMX	✓	✓	96.56	85.28	97.80	95.20	97.36	98.55	91.24	98.47	98.13	98.89	94.44	97.92	97.08
FTransUNet	✓	✓	90.35	71.52	94.14	84.62	88.16	94.65	79.25	97.83	93.11	90.22	85.76	94.16	91.80
ASANet	✓	✓	96.56	85.80	97.91	95.13	97.31	98.31	93.67	98.75	97.37	98.08	94.54	97.95	97.12
WHFNet	✓	✓	97.12	87.95	98.16	95.91	98.16	98.66	94.98	98.76	97.75	99.20	**95.46**	**98.27**	**97.58**

achieving 95.46% mIoU and 97.58% Kappa. Specifically, in terms of mIoU, it exceeds the UperNet with single optical input by 10.3% and the SAR input by 16.67%, clearly demonstrating the effectiveness and necessity of multisource data fusion. CMFNet, on the other hand, exhibits a considerable performance drop due to its sensitivity to noise interference, resulting in 72.73% mIoU only. In comparison with other homogeneous multimodal models such as MCANet, CMX, and ASANet, the proposed WHFNet achieves the highest performance across all categories in both IoU and accuracy metrics. This improvement is attributed to the use of a heterogeneous dual-branch backbone, which more effectively captures and integrates modality-specific features, thereby enhancing the semantic representation of heterogeneous data.

Furthermore, compared with SA-Gate, WHFNet achieves notable improvements in both road and greenery categories, achieving gains of 10.65% and 4.37% on the category IoU metric, respectively. These results highlight the advantage of designing fusion mechanisms that are guided by intrinsic data characteristics, which can better compensate for modality discrepancies and improve category accuracy. In addition, frequency-domain information plays a critical role in segmenting objects with complex textures or blurred structural boundaries in RS imagery. By incorporating the proposed interactor, WHFNet enhances the feature representation of edge regions for categories such as roads, buildings, and greenery. This significantly improves the model's capability to handle complex classification tasks.

Fig. 6 presents visualized classification results across six groups of diverse scenes. It can be clearly observed that WHFNet consistently achieves superior segmentation performance under various conditions, including partial and full cloud occlusion, cloud shadows, and cloud-free environments. In column (h), FTransUNet demonstrates significant misclassification between *building, greenery, and farmland* categories, particularly in the first and second groups. Due to the occlusion of thick clouds, some ground objects in the optical image are missing, which in turn affects the performance of subsequent feature fusion. In column (e), the segmentation effect is discontinuous in all areas affected by clouds, which is most obvious in the *road* category. This indicates that MCANet, due to its limited receptive fields, struggles to maintain global contextual awareness in the presence of noise and occlusions. The above situation has improved in CMX and ASANet, and no large-scale ground object misclassification is found. In column (f), although CMX effectively leverages the complementary information from SAR data through its correction and fusion mechanisms, its segmentation performance on narrow roads (groups 1 and 5) still falls short of WHFNet. In the first and sixth groups, column (g) shows that ASANet still has room for improvement in the accuracy of categories with regular geometric structures.

In contrast, WHFNet benefits from a heterogeneous architecture, enabling richer semantic feature extraction and significantly reducing misclassification. Furthermore, the integration of high-frequency information effectively mitigates the loss of edge details in the main branch, enhancing the segmentation accuracy of complex boundary regions. As illustrated in column (i), WHFNet markedly improves the continuity of road segmentation, reduces fragmentation and misclassification, and fully exploits SAR-derived structural and textural cues to generate more accurate boundary delineations.

2) Pohang Dataset: As shown in Table IV, the Pohang dataset contains thin cloud interference, enabling UperNet to achieve a high mIoU of 89.54% using optical imagery alone. Nevertheless, the proposed WHFNet demonstrates superior performance across all categories, excelling in both structurally regular objects (*buildings and roads*) and texture-rich regions (*farmland and greenery*). Notably, the most significant improvements are observed in the road and farmland categories, with mIoU gains of 2.52% and 1.34%, respectively, compared to ASANet.

Qualitative comparisons in Fig. 7 illustrate the marked enhancement of WHFNet in delineating building and water edge features (in groups 2–4). In contrast, MCANet tends to misclassify *farmland* as *greenery* under heavy fog conditions, as seen in group 4. Compared to CMX and FTransUNet, WHFNet achieves superior spatial continuity in road segmentation, benefiting from the effective fusion strategy of the interactor.

3) WHU-OPT-SAR Dataset: As presented in Table V, UperNet achieves 54.12% mIoU and 83.80% OA, surpassing the majority of existing multimodal methods. This success is largely attributed to the high quality of optical imagery in

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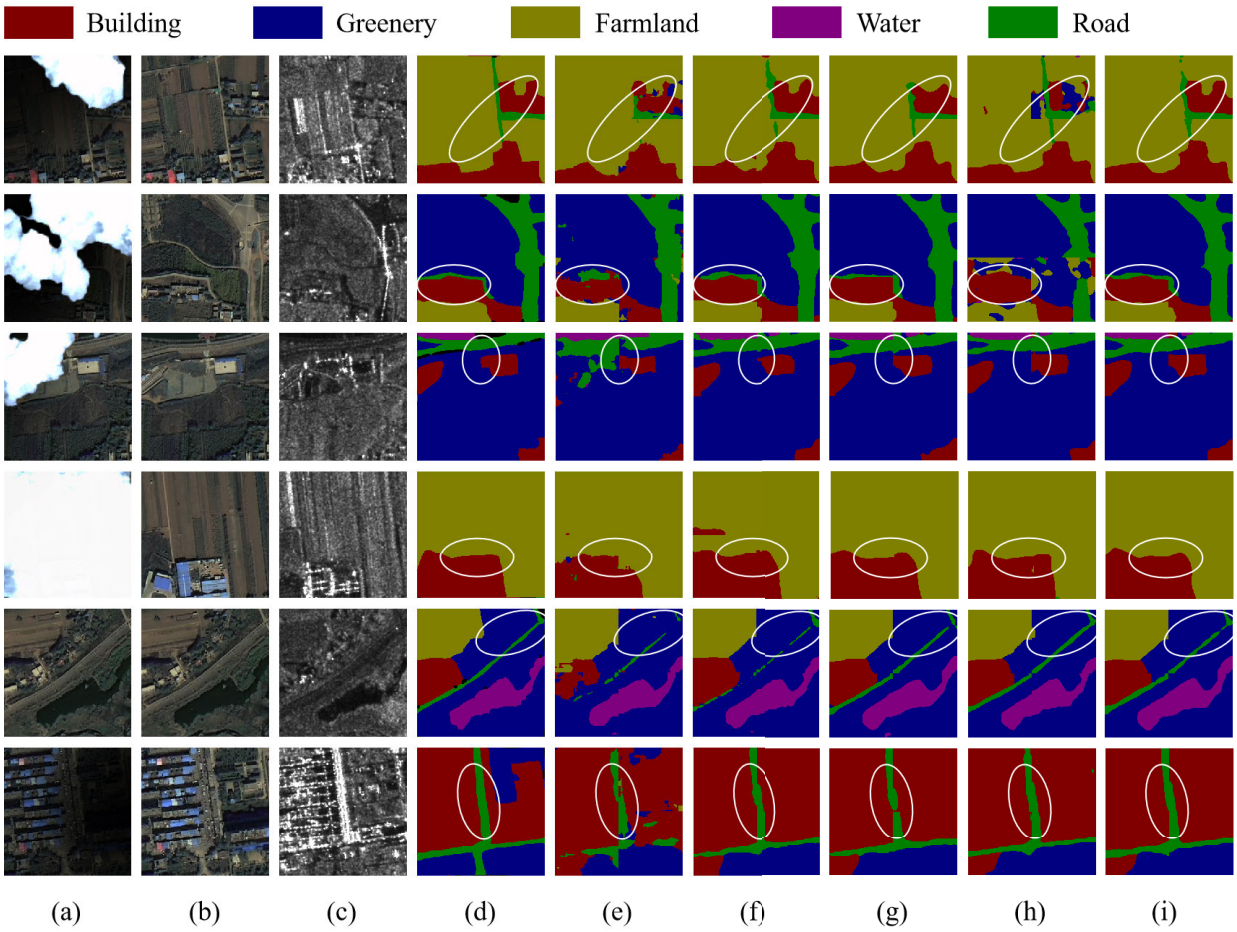


Fig. 6. LCC results of Xi'an dataset. (a) Optical images with cloud. (b) Optical images without cloud. (c) SAR images. (d) Ground truth. (e) MCANet results. (f) CMX results. (g) ASANet results. (h) FTransUNet results. (i) WHFNet results.

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON POHANG DATASET

Method	Modality		Category IoU					Category Acc					Metric		
	OPT	SAR	Building	Road	Farmland	Water	Greenery	Building	Road	Farmland	Water	Greenery	mIoU	OA	Kappa
UperNet	✓		93.79	74.68	90.69	99.03	89.48	97.39	81.73	95.10	99.33	94.35	89.54	95.95	94.26
UperNet		✓	88.80	47.81	82.04	96.83	77.51	94.75	60.61	89.88	97.79	88.46	78.60	91.50	87.96
SegFormer	✓	✓	92.77	59.94	89.13	98.69	87.87	96.77	69.53	94.49	99.23	94.20	85.68	94.85	92.71
SA-Gate	✓	✓	96.83	81.34	95.06	99.34	94.83	98.02	89.60	98.10	99.70	97.41	93.48	97.77	96.86
CMFNet	✓	✓	86.55	34.46	78.41	95.47	70.24	94.54	42.33	88.91	98.01	80.72	73.03	89.23	84.60
MCANet	✓	✓	88.25	55.05	81.88	97.74	78.60	92.21	69.50	92.84	99.26	88.24	80.30	91.69	88.33
CMX	✓	✓	97.64	87.00	96.33	99.46	96.21	98.72	92.29	98.33	99.78	98.05	95.33	98.39	97.73
FTransUNet	✓	✓	92.10	66.43	87.64	98.42	85.48	97.60	72.60	93.07	99.03	90.58	86.02	94.58	92.29
ASANet	✓	✓	97.32	85.79	95.67	99.36	95.39	98.43	92.89	98.20	99.46	97.59	94.70	98.13	97.37
WHFNet	✓	✓	97.98	88.31	97.01	99.65	96.95	99.00	93.78	98.43	99.83	98.47	**95.98**	**98.65**	**98.09**

the WHU-OPT-SAR dataset. Most approaches demonstrate strong performance in the *forest* category, where WHFNet attains an IoU of 83.69%. This is because forests exhibit distinct spectral characteristics in optical images and unique scattering signatures in SAR data, facilitating the capture of salient distinguishing features. However, *roads* present a greater challenge due to frequent occlusion by buildings and vegetation. Models such as SA-Gate and FTransUNet fail to fully exploit complementary SAR features in these scenarios, resulting in performance drops of 5.76% and 5.21% compared to UperNet (only use optical data). By contrast, WHFNet effectively integrates the long-range contextual modeling of the optical branch with the high-frequency information from DWT, achieving the highest score of 40.73%. The heterogeneous design enhances contextual pixel association in optical images and, coupled with our fusion strategy, delivers robust classification results in scenes with complex object distributions.

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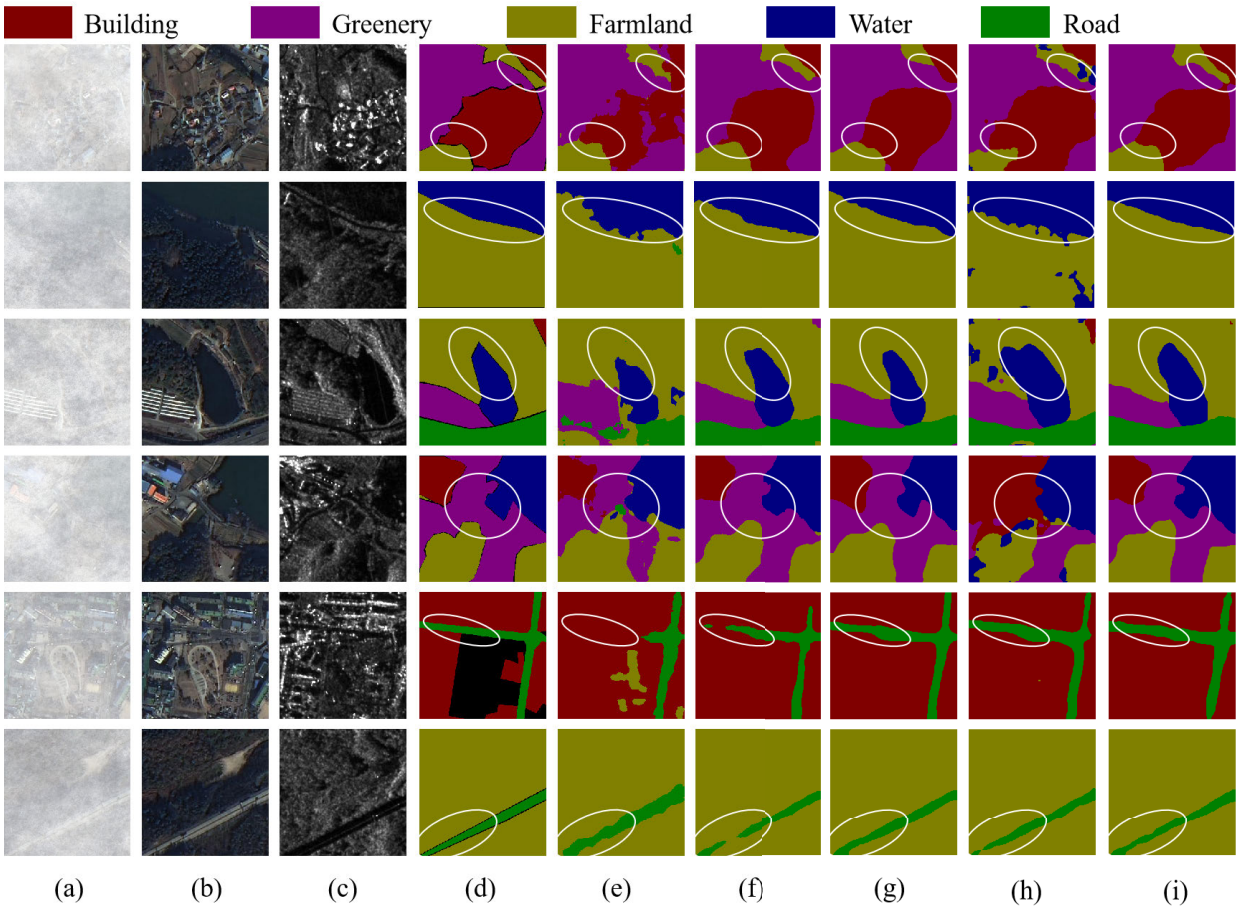


Fig. 7. LCC results of Pohang dataset. (a) Optical images with cloud. (b) Optical images without cloud. (c) SAR images. (d) Ground truth. (e) MCANet results. (f) CMX results. (g) ASANet results. (h) FTransUNet results. (i) WHFNet results.

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE WHU-OPT-SAR DATASET

Method	Modality		Category IoU							Category Acc							Metric		
	OPT	SAR	Farmland	City	Village	Water	Forest	Road	Others	Farmland	City	Village	Water	Forest	Road	Others	mIoU	OA	Kappa
UperNet	✓		68.81	55.64	48.32	63.02	83.41	38.73	20.89	80.56	74.58	62.43	76.49	92.96	55.02	25.16	54.12	83.80	75.23
UperNet		✓	60.44	52.91	31.73	54.99	78.76	22.37	14.26	74.38	74.02	44.42	62.48	92.33	29.89	21.30	45.06	78.84	67.22
SegFormer	✓	✓	66.08	51.95	42.40	59.08	81.95	23.49	18.16	79.87	77.98	57.49	69.74	91.72	30.35	22.46	49.02	81.94	72.35
SA-Gate	✓	✓	67.16	50.27	44.55	61.26	81.17	32.97	16.01	79.90	76.17	58.19	76.35	88.92	51.44	34.50	50.49	81.64	72.38
CMFNet	✓	✓	67.80	55.40	47.80	60.78	83.23	36.20	17.67	78.94	74.73	62.55	75.62	93.36	52.23	20.59	52.70	83.29	74.44
MCANet	✓	✓	66.96	49.85	46.80	58.71	82.66	33.02	13.74	79.58	73.05	62.32	75.40	91.21	47.13	17.45	50.25	82.34	73.18
CMX	✓	✓	69.34	54.07	48.54	64.31	83.49	36.61	21.71	80.69	79.59	64.12	75.16	93.11	47.27	26.85	54.01	83.98	75.53
FTransUNet	✓	✓	68.02	53.16	47.14	60.76	82.73	33.52	20.78	81.76	67.59	60.85	74.28	91.29	43.31	31.97	52.30	82.95	73.99
ASANet	✓	✓	66.19	49.02	42.33	58.09	82.31	25.55	15.20	79.90	78.17	58.23	68.44	91.76	33.83	18.86	48.39	81.84	72.24
WHFNet	✓	✓	69.85	56.30	48.36	65.21	83.69	40.73	23.45	81.26	78.09	60.79	77.90	93.04	56.46	30.25	**55.37**	**84.32**	**76.04**

Fig. 8 illustrates the segmentation results. WHFNet demonstrates superior capabilities by thoroughly leveraging the characteristics of water in SAR images (groups 1 and 4). By utilizing high-frequency information via the wavelet transform and CMI for cross-modal fusion, our model ensures the continuity of classification to a certain extent. Even for water with winding morphologies, WHFNet's predictions remain continuous and coherent. Conversely, other algorithms either misclassify *water* as *farmland* (as seen in MCANet and CMX) or identify only small area of water (in ASANet and FTransUNet). Additionally, group 5 shows that ASANet performs inadequately in the *village* category with substantial misclassifications. Regarding the *road* category, WHFNet produces better spatial coherence compared to CMX, as illustrated in the second group.

4) PIE-RGB-SAR Dataset: Table VI presents the comparative performance evaluation on the PIE-RGB-SAR dataset. It is worth noting that this dataset includes a substantial "others" class (8%). When relying solely on SAR data, UperNet

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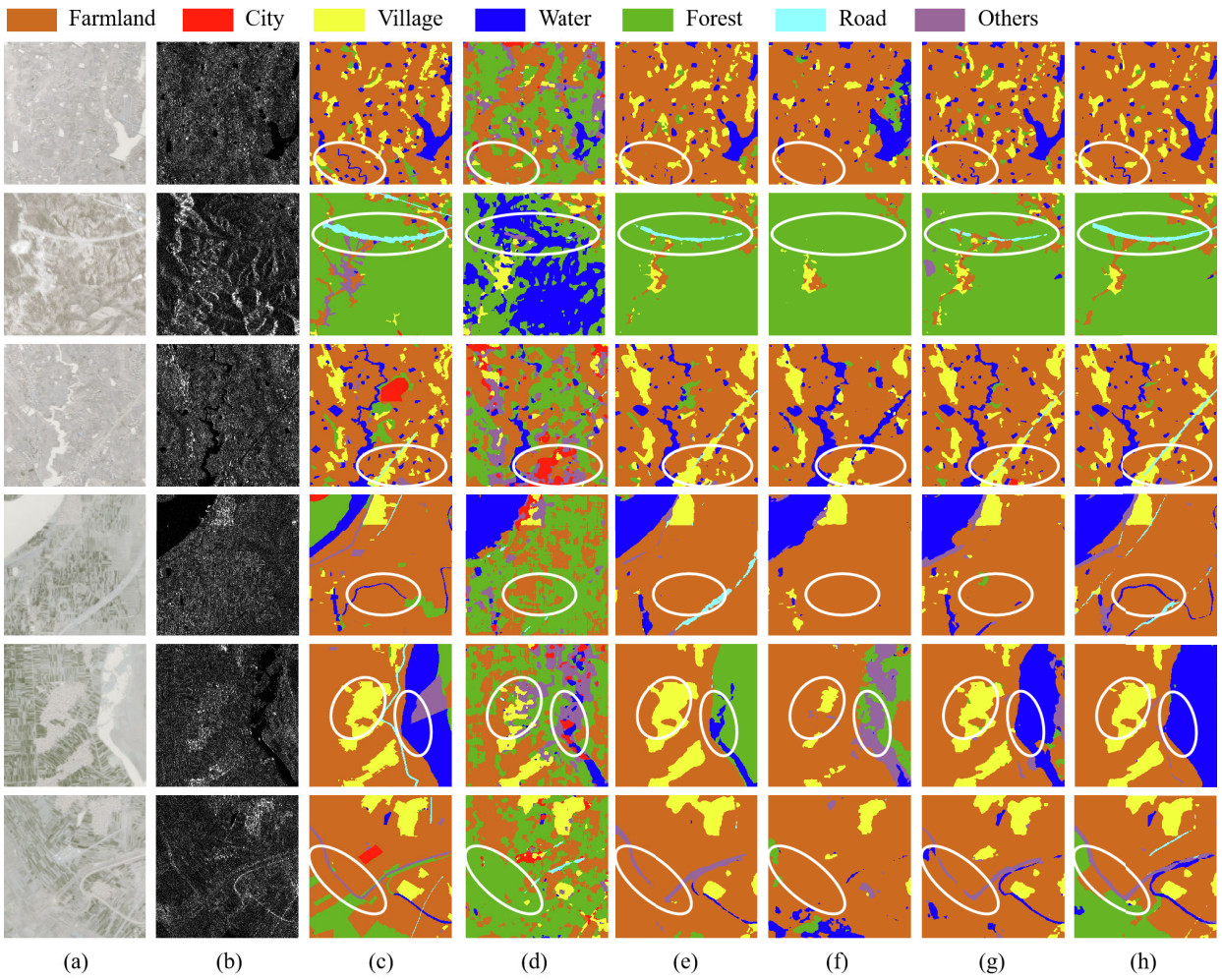


Fig. 8. LCC results of WHU-OPT-SAR dataset. (a) Optical images. (b) SAR images. (c) Ground truth. (d) MCANet results. (e) CMX results. (f) ASANet results. (g) FTransUNet results. (h) WHFNet results.

TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE PIE-RGB-SAR DATASET

Method	Modality		Category IoU						Category Acc						Metric		
	OPT	SAR	Others	City	Road	Water	Forest	Farmland	Others	City	Road	Water	Forest	Farmland	mIoU	OA	Kappa
UperNet	✓		97.11	81.32	56.70	72.70	81.33	62.95	97.85	90.75	69.36	80.77	90.91	73.85	75.35	83.92	83.25
UperNet		✓	95.92	74.67	30.61	69.00	74.34	45.26	98.09	88.11	39.97	78.79	85.96	59.02	64.97	82.80	75.49
SegFormer	✓	✓	97.07	81.24	50.29	75.13	81.60	61.22	97.97	91.71	61.43	84.46	89.71	74.56	74.43	88.09	83.13
SA-Gate	✓	✓	97.27	81.90	54.35	75.70	81.72	60.20	97.68	92.05	64.83	81.14	91.18	71.44	75.19	88.36	83.40
CMFNet	✓	✓	97.30	80.89	49.89	74.98	81.12	59.53	97.94	91.51	60.53	82.30	90.03	72.65	73.95	87.79	82.65
MCANet	✓	✓	97.23	81.22	51.43	76.31	81.39	59.47	98.08	90.36	67.88	85.48	91.29	68.11	74.51	88.04	82.97
CMX	✓	✓	97.26	82.94	56.03	77.80	83.04	63.63	98.02	92.18	68.03	86.87	91.55	72.92	76.78	89.26	84.74
FTransUNet	✓	✓	97.16	80.00	47.04	73.37	78.91	51.54	97.57	91.49	56.00	88.20	89.01	60.33	71.33	86.43	80.67
ASANet	✓	✓	97.18	83.41	61.20	79.79	83.45	65.08	98.19	91.30	76.57	87.45	92.24	74.06	78.35	89.74	85.42
WHFNet	✓	✓	97.52	83.45	60.67	79.61	83.74	65.47	98.14	92.10	72.86	87.55	92.20	74.20	**78.41**	**89.86**	**85.59**

underperforms compared to other methods, particularly in the *water* and *others* categories. Due to the inherent imaging characteristics of SAR, water typically exhibits specular reflection, resulting in weak backscattering signals and dark pixel intensities. Consequently, they are prone to being misclassified as the similarly dark *"others."* However, the integration of optical data stabilizes performance in methods like SA-Gate and MCANet, underscoring the vital complementarity between optical and SAR modalities for the LCC task. The proposed method achieves the highest overall performance, with an mIoU of 78.41% and an OA of 89.86%. These superior metrics confirm that the heterogeneous network architecture enhances feature representation. Furthermore, our method shows better performance in the *forest* and *city* categories, which constitute

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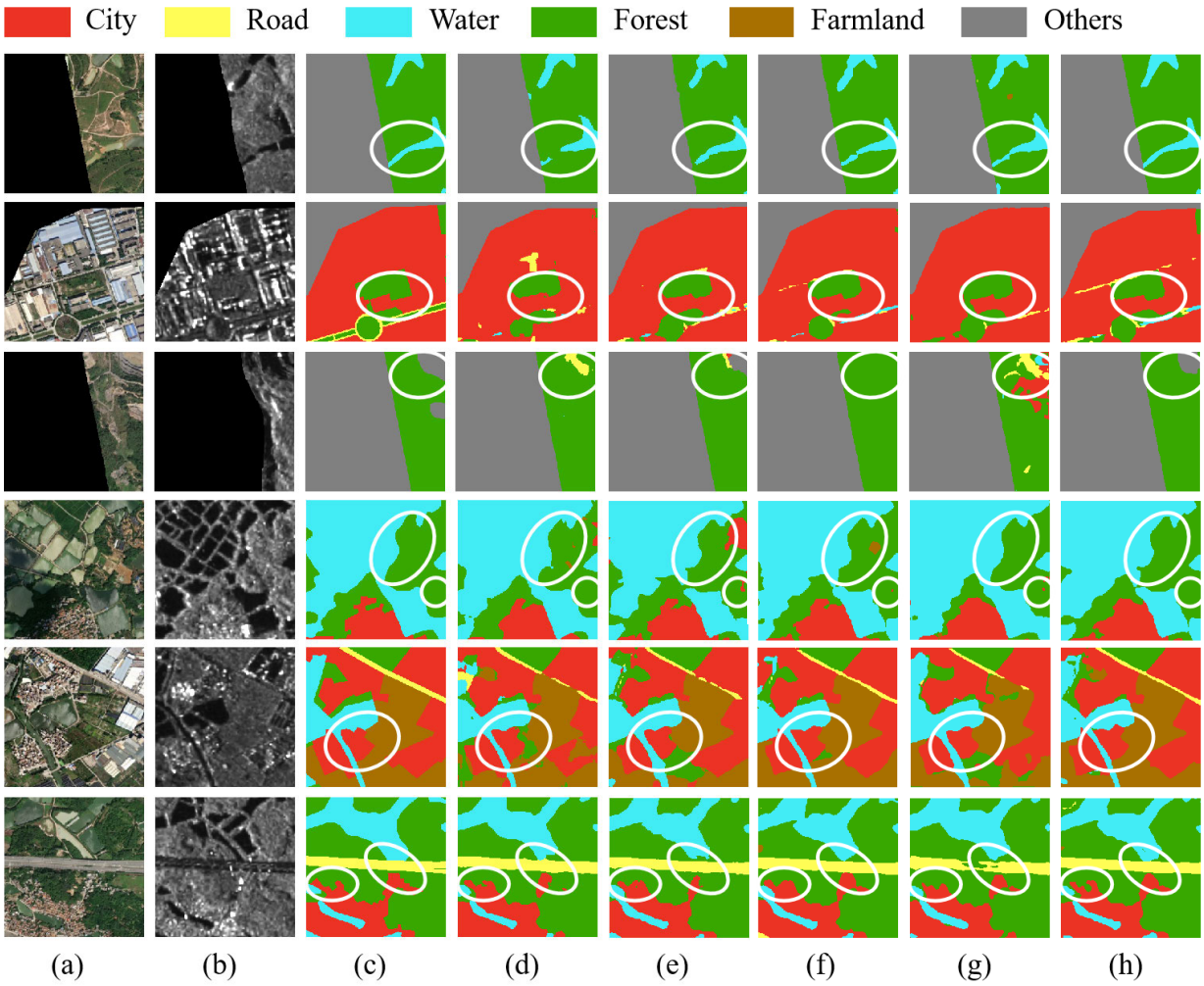


Fig. 9. LCC results of PIE-RGB-SAR dataset. (a) Optical images. (b) SAR images. (c) Ground truth. (d) MCANet results. (e) CMX results. (f) ASANet results. (g) FTransUNet results. (h) WHFNet results.

a significant portion of the dataset, highlighting its capabilities in large-scale scene modeling.

The visualization results in Fig. 9 further corroborate these findings. Leveraging high-frequency information from the wavelet transform, WHFNet generates smoother boundaries in the predictions for the city and forest regions (see column h). Other methods, including CMX, MCANet, and ASANet, suffer from misclassification, such as mislabeling portions of large *forest* areas as *city* or *farmland*. In contrast, WHFNet produces cleaner segmentation maps. As shown in groups 1–4, our effective fusion of spatial and frequency features results in outputs with minimal classification noise and high semantic consistency.

D. Ablation Experiment

To validate the improvement of the different components, step-by-step ablation studies were conducted on the Xi'an and Pohang datasets, with results detailed in Table VII. The experimental configurations include: Swin-OPT denotes single-modal optical input, ConvNeXt-SAR refers to single-modal SAR input, HF indicates the incorporation of high-frequency information with the additive fusion strategy, SFFM represents the proposed SFFM, and SCL stands for the structural consistency loss.

1) Effectiveness of Components: The heterogeneous dual-branch architecture significantly outperforms single-modal inputs, demonstrating performance gains exceeding 10% on both datasets. This highlights the substantial potential of heterogeneous backbone fusion for multimodal RS data. While the introduction of HF enhances road segmentation details in the Pohang dataset, where road class accuracy improves by 1.53%. However, in the Xi'an dataset, which contains thick cloud, the retention of high-frequency information at cloud edges leads to a slight water category IoU reduction, directly motivating the necessity of our spatial–frequency fusion strategy. The SFFM module effectively captures complementary spatial- and frequency-domain differences through subtraction operations, simultaneously mitigating noise propagation. This results in remarkable performance improvements, achieving 95.20% mIoU for Xi'an and 95.67% for Pohang, underscoring the criticality of an appropriate fusion method for effective representation learning. Additionally, using HF and SFFM improves the baseline (heterogeneous dual-branch, third row)

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TABLE VII

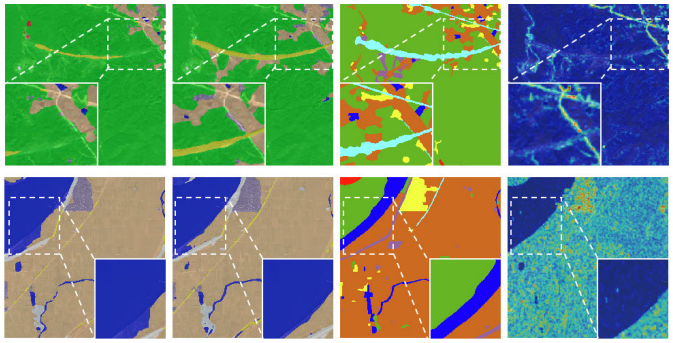
ABLATION ON EFFECTIVENESS OF VARIOUS COMPONENTS ON XI'AN AND POHANG DATASETS

Dataset	Components					Category IoU					Metric		
	Swin-OPT	ConvNext-SAR	HF	SFFM	SCL	Building	Road	Farmland	Greenery	Water	mIoU	OA	Kappa
Xi'an	✓					85.86	54.65	90.89	78.19	90.88	80.10	91.23	87.67
		✓				86.47	58.02	90.39	76.83	89.01	80.14	91.19	87.62
	✓	✓				95.22	81.49	97.02	93.30	97.11	92.83	97.17	96.03
	✓	✓	✓			95.35	81.48	97.04	93.39	97.07	92.87	97.21	95.97
	✓	✓	✓	✓		96.99	87.19	98.04	95.76	98.02	95.20	98.18	97.44
	✓	✓	✓	✓	✓	97.12	87.95	98.16	95.91	98.16	**95.46**	**98.27**	**97.58**
Pohang	✓					88.02	48.92	82.72	81.85	98.13	79.93	92.01	88.61
		✓				91.09	58.95	85.27	81.80	97.88	83.00	93.39	90.64
	✓	✓				96.73	80.5	95.18	94.91	99.46	93.36	97.76	96.84
	✓	✓	✓			96.90	82.03	95.31	95.05	99.48	93.75	97.87	96.99
	✓	✓	✓	✓		97.85	87.38	96.82	96.70	99.63	95.67	98.55	97.95
	✓	✓	✓	✓	✓	97.98	88.31	97.01	96.95	99.65	**95.98**	**98.68**	**98.09**

TABLE VIII

COMPARISON OF DIFFERENT WAVELET BASES ON XI'AN DATASET

Wavelet	mIoU↑	mAcc↑	mDice↑	Kappa↑
Haar	**95.40**	**97.87**	**97.60**	**97.54**
Db2	95.20	97.76	97.50	97.44
Bior2.4	95.16	97.75	97.48	97.42
Sym4	95.12	97.69	97.45	97.39



(a) (b) (c) (d)

Fig. 10. Visual analysis of high-frequency features. The high-frequency components highlight the detailed information to improve the coherence and continuity of specific category classifications, such as water and roads. (a) w/o HF: results without HF features. (b) +HF input: results with the addition of HF feature input. (c) GT and our prediction map. (d) Visualization showing the attention of HF components.

by 2.37% in Xi'an. The subsequent integration of the SCL loss optimizes category boundaries, particularly for buildings and roads. These results conclusively demonstrate the synergistic benefits of wavelet high-frequency information and cross-modal fusion strategy within the proposed heterogeneous optical–SAR framework.

2) Comparison of Wavelet Bases: Table VIII additionally compares various wavelet bases, including Haar, Daubechies 2 (Db2), Biorthogonal 2.4 (Bior2.4), and Symlets 4 (Sym4) on the Xi'an dataset. Our analysis indicates that the Haar wavelet offers superior pixel-wise accuracy and segmentation performance while maintaining computational efficiency. In Section IV-C, we apply the Haar wavelet as a wavelet basis.

TABLE IX

IMPACT OF COMPONENT SELECTION ON FUSION STRATEGY PERFORMANCE

Fusion Strategy	Xian		WHU	
	mIoU ↑	mDice↑	mIoU↑	mDice↑
LF	95.28	97.54	55.28	69.17
HF+LF	95.27	97.54	55.12	69.04
HF(ours)	**95.46**	**97.64**	**55.37**	**69.35**

3) Qualitative Results: The analysis of frequency components is presented in Fig. 10 and Table IX. Notably, relying solely on the main feature extraction pathway [Fig. 10(a)] often results in the loss of fine boundary details. In contrast, the high-frequency components generated by the wavelet transform enrich the feature representation, thereby facilitating more efficient downstream processing within the CMI. As shown in Fig. 10(d), this enhanced high-frequency information enables the interactor to better fuse cross-modal features, yielding more precise segmentation. Table IX further supports this observation. The HF-only strategy achieves the best performance on both Xi'an and WHU-OPT-SAR datasets (95.46% and 55.37% mIoU, respectively), while incorporating the full frequency components (HF+LF) leads to a slight performance decline. This indicates that LF components may introduce redundancy and amplify SAR speckle noise.

As a critical component of the CMI, the SFFM is designed to capture the distinctions between frequency and spatial domain features, guiding the model to focus on key regions for discriminative representation learning. Unlike standard convolutional layers, the SFFM incorporates an adaptive mapping function and an attention mechanism, demonstrating strong cross-modal capability. To verify its effectiveness, we provide a visual comparison of the intermediate feature representations in Fig. 11. Columns (a) and (b) display the input images and backbone features, respectively. Column (c) illustrates the high-frequency features from the texture-aware block, while (d) displays the fused features after applying the SFFM. Evidently, the fused features not only preserve the semantic abstraction from the backbone but also integrate detailed information from HF, resulting in enhanced spatial coherence

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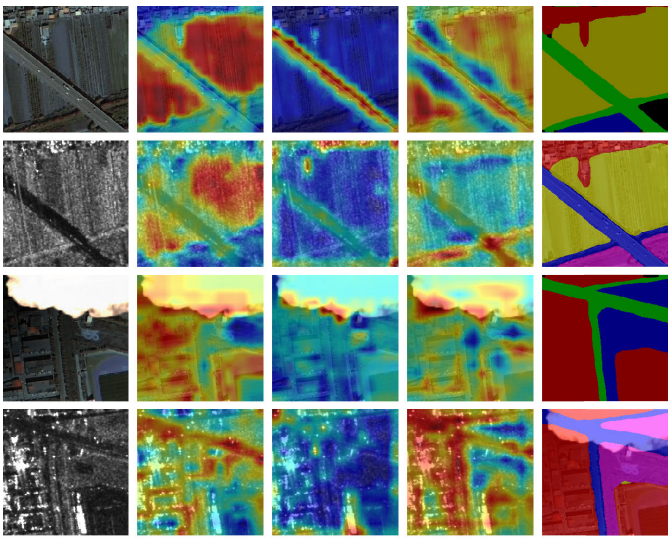


Fig. 11. Visualization of spatial–frequency fusion. (a) Input images. (b) Backbone features. (c) High-frequency features from the texture-aware block. (d) Fused features after SFFM. (e) Ground truth and predictions of our method.

and sharper object boundaries. Finally, column (e) shows the ground truth and predictions of our method. There is no doubt that WHFNet achieves highly accurate and structurally consistent predictions.

V. Conclusion

In this article, we propose a heterogeneous cross-modal fusion network (WHFNet) for optical–SAR LCC. By structurally decoupling feature encoding and explicitly modeling high-frequency details via wavelet interactions, the proposed WHFNet effectively improves the classification performance across diverse resolutions and complex scenarios. In addition, a unified module CMI is designed to fuse cross-modal features. Furthermore, the SCL is developed to regularize intermodal representation alignment. Extensive experiments conducted on four multimodal RS datasets demonstrate the effectiveness of our method. In the future, we plan to explore adaptive frequency decomposition strategies and enhance robustness to sensor misalignment and domain shifts in real-world applications.

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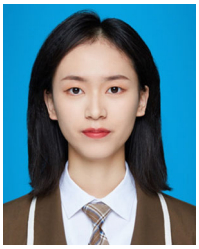
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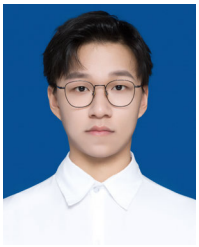
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